

GT2026-176436

DRAFT: UNAVAILABLE ENERGY FRAMEWORKS FOR AVIATION HEAT EXCHANGER DESIGN

Kilian P. Bartsch^{1,*}, Zelimhan Raduev¹, Robert J. Miller¹, Nicholas R. Atkins¹,

¹Whittle Laboratory, University of Cambridge, Cambridge, United Kingdom

ABSTRACT

This paper compares frameworks for trading heat transfer and pressure drop in mass-constrained heat exchangers for aviation applications. Rather than working with effectiveness and pressure drop separately, two frameworks assessing the value of heat will be compared. It will be shown that the classical availability framing (exergy), applied naively to the heat exchanger in isolation, suggests that no heat transfer is always better. The physical insight being that heat transfer across a temperature difference is detrimental for classical available energy. This work then applies the practical availability framework (euergy), which captures the right physical mechanism and is a convenient way for heat exchanger designers to trade heat transfer and pressure drop without detailed knowledge of the cycle in which the performance-enhancing heat exchanger is used. This framing suggests contrarily that large temperature differences, rather than being detrimental, are a helpful lever to increase the value of heat without excessive viscous dissipation (pressure drop). The practical availability framing is then successfully applied to improve a generic helicopter recuperator based on a high-fidelity design from literature. Three first order heat exchanger design choices will be studied via a general heat exchanger scaling model. These design choices involve trade-offs between heat transfer and pressure drop. The practical availability framework will be shown to help determine first order the general direction of design for mass and volume limited heat exchangers.

Keywords: Heat Exchangers, Work Cycles, Entropy Production, Euergy, Exergy

1. INTRODUCTION

The transition toward cryogenic hydrogen or methane as an aviation fuel enhances the viability of advanced engine cycles that utilise significant heat exchange. Integrating these heat exchangers introduces a critical trade-off between heat transfer and pressure loss. Given stringent weight and volume constraints, optimising heat exchanger geometry for a specific cycle remains a

complex multi-dimensional design challenge. Traditionally, heat exchangers are designed by calculating pressure drop and heat transfer in isolation before integrating them into a cycle to assess performance. This process is then repeated and the final design achieved through iteration. This approach is fundamentally limited; because the design space is highly complex, the designer often lacks the physical intuition required to understand how specific geometric changes will impact overall cycle efficiency.

An alternative approach is to employ a classical availability (exergy) analysis [1]. As illustrated in Fig. 1a, classical availability represents the maximum work extractable from a flow by a universal reversible machine, comprising a reversible turbine that expands the flow to ambient pressure, followed by a reversible heat engine that cools it to the reference temperature. The performance of the heat exchanger is then the change of classical availability across the heat exchanger.

Classical availability establishes a direct physical link between irreversibilities within the heat exchanger and the resulting

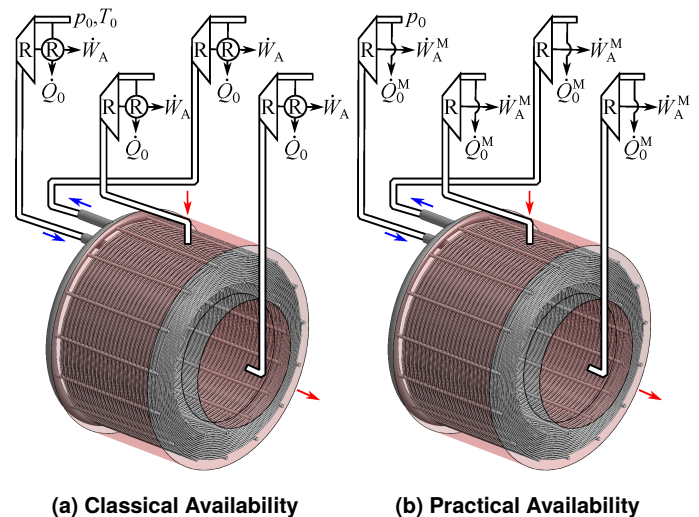


FIGURE 1: Two availability methods for heat exchanger analysis

*Corresponding author: kpb30@cam.ac.uk
Draft Version 8., February 11, 2026.

drop in its performance. While this provides insight into the location of losses, it can be functionally misleading during the design process. Because any irreversibility, whether thermal or viscous, reduces the availability across the component, which implies that the optimal solution is to remove the heat exchanger entirely. However, at the cycle level, the heat transfer often improves the overall efficiency. This inherent contradiction has led many in the industry to reject classical availability in favour of conventional iterative methods of design, as the former fails to intuitively account for the system-wide advantages of the component.

To resolve this, this paper introduces an alternative method: practical availability (eergy), as developed by Miller [2]. As illustrated in Fig. 1b, practical availability is defined as the maximum work extractable from a flow using a reversible turbine alone to reach ambient pressure. Unlike classical availability, no reversible heat engine is employed; any remaining work potential being lost in the exhaust of the ideal engine. The performance of the heat exchanger is then given by the change in practical availability across the heat exchanger.

Practical availability is particularly effective for analysing flows with simultaneous thermal and viscous irreversibilities, such as cooled turbines, Jardine and Miller [3]. Like the classical approach, it establishes a direct physical link between specific internal irreversibilities and performance change. However, it will be shown that because the ideal work extraction process and the real work extraction process are aligned, changes in the performance of the heat exchanger align with changes of the performance of the whole cycle.

The paper is divided into four primary sections. The first compares three analytical frameworks—conventional, classical availability, and practical availability to the problem of determining the best length of a heat exchanger for a fixed frontal area. The reader will learn that the practical availability method enables answering the question of whether a heat exchanger with effectiveness $\varepsilon = 60\%$ and pressure drops $\Sigma_i(\Delta p/p)_i = 12\%$ is better than a heat exchanger with effectiveness $\varepsilon = 54\%$ and pressure drops $\Sigma_i(\Delta p/p)_i = 5\%$ for work extraction. Thereafter, the heat exchanger model used for scaling heat exchangers in mass constrained applications like aviation will be presented. Finally, a helicopter engine recuperator design is presented to demonstrate the practical utility of the practical availability method for designers.

2. COMPARISON OF FRAMEWORKS

In this section the three frameworks are applied to the design choice of heat exchanger length for a fixed free flow area and hence mass velocity G . Three heat exchangers are considered, each considered to have a different dimensionless mass velocities g^2 [4].

$$g_i^2 \triangleq \underbrace{\left(\frac{\dot{m}_i}{A_{o,i}}\right)^2}_{G_i^2} \frac{1}{2\rho_i p_{i,\text{in}}} = \frac{\overline{\rho_i v_i^2}}{2p_{i,\text{in}}} \quad \text{for } i \in \{\text{hot, cold}\} \quad (1)$$

which can also be thought of as an average bulk dynamic pressure divided by the inlet static pressure p_{in} . The mass velocity

is commonly used in heat exchanger design as it remains constant even as the density changes due to heat transfer. ρ is a suitably chosen average density. In this work, the focus is on the magnitude of g^2 and hence it will be approximated by the inlet density.

For each design the length is varied until the optimal length was achieved. The results of the three framework analyses is shown in Figure 2.

2.1. Conventional Framework

The variation in heat transfer and pressure drop for the three designs of heat exchanger are shown in Fig. 2a. As the length of the heat exchanger is increased the heat transfer effectiveness rapidly increases before starting to plateau. The pressure drop rises linearly, the gradient of which is dependent on the magnitude of the dimensionless mass velocity.

It can be seen that no clear optimal exists and designers often decide on a percentage pressure loss that can be tolerated and then attempt to maximise the heat transfer effectiveness. Alternatively the required heat transfer can be specified and the pressure drop minimised.

2.2. Classical Availability (Exergy)

In a classical availability (exergy) the unavailable energy can be changed by either heat transfer between the streams or viscous dissipation, resulting in pressure drop, in either stream. The change in unavailable energy is given by:

$$\Delta\dot{Q}_0 \triangleq T_0 \sum_{i \in \{\text{hot, cold}\}} \dot{m}_i (s_{i,\text{out}} - s_{i,\text{in}}) \quad (2)$$

It becomes possible to now compare sources of irreversibility (such as heat transfer across temperature gradients and viscous dissipation) and their impact on cycle efficiency. Bejan has extensively applied this framework to heat exchanger design, and has shown that for a counterflow heat exchanger with balanced heat capacity flow rates, transferring heat between two ideal gases, the net entropy generation $\dot{S}_{\text{gen}}$ in the heat exchanger (combining both streams) can be written as a function of the heat transfer effectiveness ε and pressure drop $\Delta p/p_{\text{in}}$ and is proportional to the classical unavailable energy creation as defined in (2):

$$\frac{\dot{S}_{\text{gen}}}{C_{\text{min}}} = \frac{\Delta\dot{Q}_0}{\dot{Q}_{\text{max}}} \cdot \frac{T_{\text{hot,in}} - T_{\text{cold,in}}}{T_0} \quad (3)$$

$$= \ln \left[1 + \frac{(\mathcal{T} - 1)^2}{\mathcal{T}} \cdot \varepsilon(1 - \varepsilon) \right] - 2k \ln \left(1 - \frac{\Delta p}{p_{\text{in}}} \right) \quad (4)$$

where the heat capacity flow rate $C_i = \dot{m}_i c_{p,i}$ for stream i , $C_{\text{min}} = \min_i(C_i)$, $\mathcal{T} = T_{\text{hot,in}}/T_{\text{cold,in}}$ is the inlet temperature ratio, $k = 1 - 1/\gamma$ is the isentropic exponent and γ is the heat capacity ratio for the streams.

Equation (4) is valid for $C_r = 1$ and equal pressure drops for both fluids (of equal heat capacity ratio and hence equal k).

When applied to the heat exchanger the change in unavailable energy of the heat exchanger is shown in Fig. 2b. The plots show a several things. First the lowest unavailable energy always occurs when the number of heat transfer units are zero. This is because

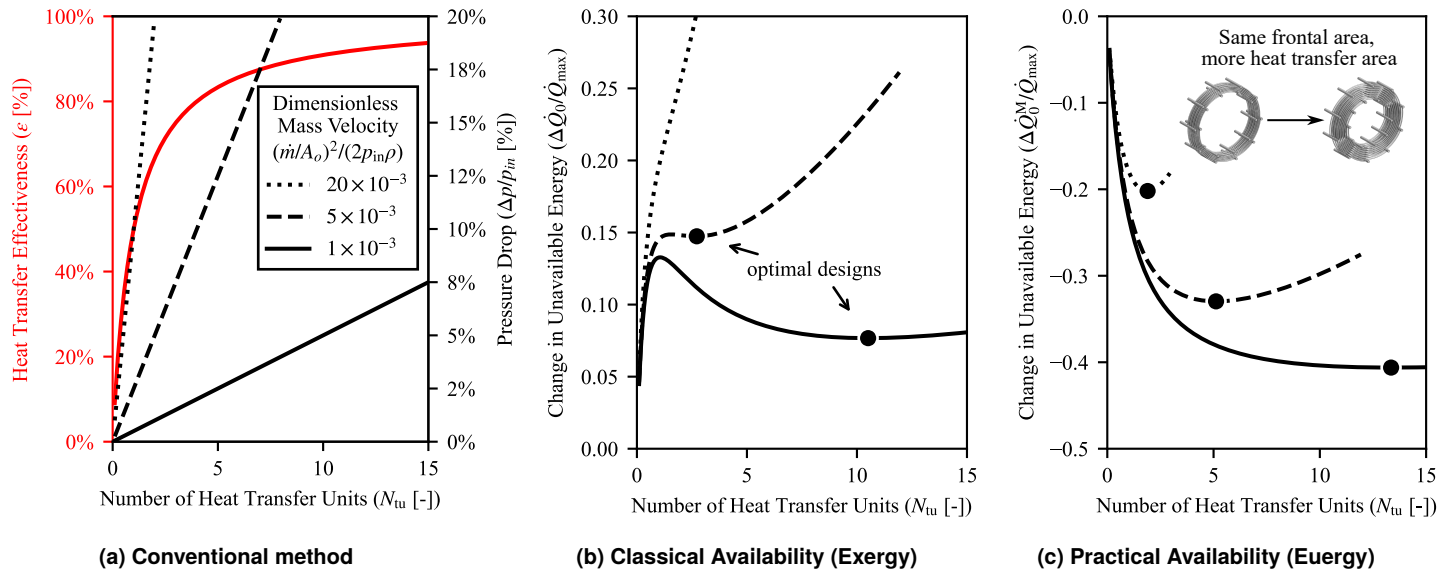


FIGURE 2: A comparison of three frameworks for a heat exchanger as its length is varied. Three heat exchangers of different dimensionless mass velocities are considered. The mass and the volume of the heat exchanger are unconstrained as the length is increased.¹

TABLE 1: g^2 for different HEx designs and concepts.

Recuperator	G_{hot} [kg m ⁻² s ⁻¹]	g_{hot}^2 [-]	Source
CCGT HRSG	6	9×10^{-4}	[5]
WR-21	22	1×10^{-2}	[6]
LH ₂ aircraft	23	7×10^{-2}	[7]
Helicopter ²	96	8×10^{-2}	[8]
Helicopter ³	55	2×10^{-2}	[8]

both terms above are positive and therefore this framework always shows that the best heat exchanger is the one that does not exist.

Second, for the two heat exchangers with lower dimensionless mass velocity a local minimum does exist. This is not the global minima but could be considers some form of optimal design. The third heat exchangers with highest dimensionless mass velocity does not have a local minimum.

It is clear from these plots that classical availability tells you that the heat exchanger with the lowest unavailable energy is the heat exchanger which does not exist. This clearly demonstrates that classical availability is not a good way of analysing the performance of a heat exchanger.

Given the assumptions in Fig. 2, Assuming balanced heat exchanger, balanced Stanton numbers and equal hydraulic diameters, the threshold g^2 for minimum entropy generation derived by the authors, more details in Appendix A.

$$g^2 > 0.05 \cdot \frac{St/f}{(1 + d_{h,\text{cold}}/d_{h,\text{hot}})} \left(\frac{T_{\text{hot,in}}}{T_{\text{cold,in}}} - 1 \right) = 0.01 \quad (5)$$

Table 1 illustrates the wide range of g^2 values across different applications. The WR-21 marine engine and the LH₂ aircraft recuperator have similar dimensional mass velocities ($G_{\text{hot}} \approx 22\text{--}23 \text{ kg m}^{-2} \text{ s}^{-1}$), yet the dimensionless g^2 differs by a factor of seven. This discrepancy arises from different operating conditions: the WR-21 operates at sea level with higher air density, while the LH₂ aircraft recuperator operates at cruise altitude where reduced air density results in much higher g^2 despite similar dimensional velocities. Consequently, the LH₂ aircraft case exhibits g^2 values comparable to the helicopter recuperator, reflecting the constraints of aviation applications where frontal area is severely limited.

2.3. Practical Availability (Eurgy)

In practical availability (eurgy) the unavailable energy of can be changed by either heat transfer between the stream or viscous dissipation, resulting in pressure drop, in either stream.

Assuming perfect gas behaviour (ideal gas and constant heat capacity with temperature), the net change in practical unavailable energy of both streams of a heat exchanger is given by:

$$\Delta \dot{Q}_0^M \triangleq \sum_{i \in \{h,c\}} \dot{m}_i c_{p,i} \left[T_{i,\text{out}} \left(\frac{p_0}{p_{i,\text{out}}} \right)^{k_i} - T_{i,\text{in}} \left(\frac{p_0}{p_{i,\text{in}}} \right)^{k_i} \right] \quad (6)$$

where $k_i = 1 - 1/\gamma_i$ is the isentropic exponent and γ_i is the heat capacity ratio for stream i . Throughout this work, all temperatures and pressures considered are the true thermodynamic static properties of the fluid, and it is assumed that there is negligible change in kinetic energy for each stream.

¹ Assumes equal heat capacity flow rates and that hot side pressure drop (plotted) is much greater than cold side relative to inlet pressures. Assume $St/f = 0.4$ for both fluids, equal friction factors $f_{\text{hot}} = f_{\text{cold}}$, equal hydraulic diameters $d_{h,\text{hot}} = d_{h,\text{cold}}$ and that the pressure drop is dominated by the core friction component (neglects entry, exit and expansion pressure changes).

² Retrofit of heat exchanger bolted-on to a largely unmodified engine

³ Design of bespoke engine for heat exchanger

For a heat exchanger between two perfect gases of equal γ , $\dot{m}$ and c_p with small relative pressure drops, such that $(1 - \Delta p/p_{\text{in}})^{-k} \approx 1 + k\Delta p/p_{\text{in}}$ and higher order pressure drop terms are neglected, the net change in unavailable energy can be linearised as follows to obtain an expression based on the thermal effectiveness ε and the $\dot{P}_i = \dot{m}_i \Delta p_i / \rho_i$ is the pumping power for stream i :

$$\varepsilon_M \approx \underbrace{\left[\left(\frac{p_0}{p_h} \right)^k - \left(\frac{p_0}{p_c} \right)^k \right]}_{\text{Net thermal creation}} \cdot \varepsilon - \underbrace{\left(\frac{p_0}{p_h} \right)^k \frac{\dot{P}_h}{\dot{Q}_{\text{max}}} - \left(\frac{p_0}{p_c} \right)^k \frac{\dot{P}_c}{\dot{Q}_{\text{max}}}}_{\text{Viscous dissipation}} \quad (7)$$

where the practical availability or mechanical work potential effectiveness of the heat exchanger ε_M is the net change in practical unavailable energy across the heat exchanger for both streams, normalised by the maximum possible heat transfer:

$$\varepsilon_M \triangleq \frac{-\Delta \dot{Q}_0^M}{\dot{Q}_{\text{max}}} \quad (8)$$

The first term represents the thermal creation term, as opposed to the classical availability framework where there is only thermal dissipation from heat transfer across temperature differences, the practical availability framework's thermal term can be negative for two streams if heat is transferred from a low-pressure stream to the high-pressure stream: $p_c > p_h$. In recuperators, the bracketed term is positive, indicating that by attributing the right value to heat, net work potential creation. The remaining terms represent the penalty of viscous dissipation, weighted by the pressure ratio of each stream.

For a heat exchanger with small relative pressure drops and both streams behaving as perfect gases with similar heat capacities, the ratio of pumping power to maximum heat transfer can furthermore be approximated as:

$$\frac{\dot{P}}{\dot{Q}_{\text{max}}} \approx k \frac{\Delta p/p_{\text{in}}}{1 - T_{c,\text{in}}/T_{h,\text{in}}} \quad (9)$$

This ratio equals the relative pressure drop $\Delta p/p_{\text{in}}$ when $T_{h,\text{in}}/T_{c,\text{in}} = \gamma$, which is often approximately satisfied in recuperators (e.g., for air with $\gamma \approx 1.4$ and typical inlet temperature ratios of 1.3–1.5).

This formulation reveals that the value of heat transfer relative to pressure drop depends on the pressure levels of each fluid, not merely on the total entropy generated. A heat exchanger operating between streams at similar pressures will have a small benefit term, making it more sensitive to viscous dissipation penalties.

Figure 2c applies this framework to the problem of lengthening the counterflow heat exchanger, and by attributing the right value of heat relative to work, even at high dimensionless mass velocity, an optimum can be found. Indeed, rather than considering that heat transfer across temperature differences is a source of irreversibility that must be minimized, this framework captures the asymmetry between the ease of extracting mechanical work with turbines, versus the difficulty of extracting thermal work potential with reversible heat engines. Therefore, by transferring

heat from a stream at lower pressure to a higher-pressure stream, this framework captures the avoided downstream loss by attributing the right value to heat. Now the designer must not assume knowledge of downstream processes out of their control and can proceed to making decisions that involve non-obvious trade-offs between heat transfer and pressure drop.

2.4. Method Summary

Figure 3 summarises the breakdown of energy into unavailable and available energies for the two availability frameworks. The classical availability framework considers the absolute upper bound of how much energy can be extracted from any machine. The practical availability framework on the other hand considers the energy that can be accessed by direct mechanical work extraction. Both are useful tools to designers, but the practical availability framework is particularly relevant when considering a single heat exchanger in mass constrained applications, as no bottoming cycle will be present, which could approximate the reversible heat engine present in the classical availability framework.

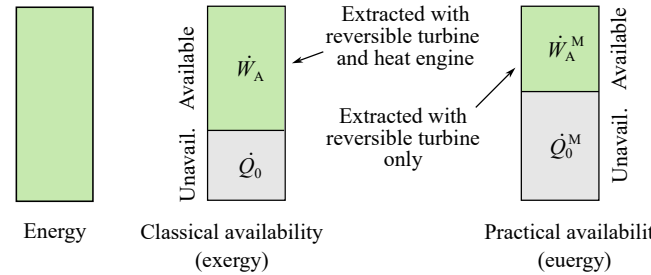


FIGURE 3: Breakdown of energy into unavailable and available energies for the two availability frameworks.

For the remainder of the paper, we now return to the specifics of aviation heat exchangers, where the mass and volume constraints make trading pressure drop and heat transfer of particular importance.

3. HEAT EXCHANGER SCALING MODEL

In this section, the equations and assumptions behind the general heat exchanger model used to demonstrate the framework of the paper are presented. The goal is to consider geometric scaling decisions that involve a trade-off between heat transfer and pressure drop, not to provide a high-fidelity model for a particular problem.

3.1. Geometric Design Parameters

3.1.1. Independent Parameters As the following case studies will be based on counterflow and (multi-pass) crossflow heat exchangers, the geometric parametrisation has been chosen to be independent of relative flow orientation of the two fluids. The primary independent parameter was chosen to be the heat transfer area A_i (including fins) and the HEx core volume occupied by the fluid, V_i , for each fluid side i . As they will be different for the two fluids, the ratios are considered independent variables so that one side's can be chosen as reference.

$$A_r \triangleq \frac{A_{\text{cold}}}{A_{\text{hot}}} \quad (10) \quad V_r \triangleq \frac{V_{\text{cold}}}{V_{\text{hot}}} \quad (11) \quad \sigma_r \triangleq \frac{A_{o,\text{cold}}}{A_{o,\text{hot}}} \quad (12)$$

For aligned flow (counter and coflow) heat exchangers the core flow lengths L_i are identical. If the free-flow areas $A_{o,i}$ are also constant $V_r = \sigma_r$. These ratios are inspired by, but slightly different to, the ratios defined in Milten et al. [9].

The area ratio A_r can primarily be leveraged using fins: if there are no fins or the surfaces are geometrically identical it is unity but can be much greater or smaller than unity to provide more heat transfer area (and friction area) to one fluid. The volume and free-flow area ratios V_r and σ_r allocates more or less of the total volume to one fluid and are the key geometric lever impacting velocity.

The relationship between area and volume is controlled by the hydraulic diameter $d_{h,i}$ (Kays and London definition [10]). To maintain a general treatment including crossflow tube banks, the true volume based hydraulic diameter $d_{v,i}$ is more convenient CITE XX.

$$d_{h,i} \triangleq \frac{4A_{o,i}L_i}{A_i} \leq d_{v,i} \triangleq \frac{4V_i}{A_i} \quad i \in \{\text{cold, hot}\} \quad (13)$$

By definition, $d_{v,\text{cold}}/d_{v,\text{hot}} = V_r/A_r$ and for identical flow lengths, $d_{h,\text{cold}}/d_{h,\text{hot}} = \sigma_r/A_r$.

For the sake of this work, the absolute values of these diameters are considered fixed to a small value limited by manufacturability and reliability constraints.

3.1.2. Quantities that scale with the heat transfer area

The HEx is assumed to have a metal or wall volume V_{metal} proportional to the heat transfer area. As the heat transfer area can potentially include fins, this assumes that the fin thickness is proportional to the main conduction thickness. The constant of proportionality is defined as follows:

$$t_{\text{hot}} \triangleq \frac{V_{\text{metal}}}{A_{\text{hot}}} \quad (14)$$

Therefore the mass of the heat exchanger core is given by the metal material density ρ_{metal} :

$$m_{\text{core}} = \rho_{\text{metal}} \cdot t_{\text{hot}} \cdot A_{\text{hot}} \quad (15)$$

The volume of the core, V_{core} , can be broken down into three volumes occupied by the metal walls and the two fluids and rearranged using the volume ratio (11) and the volume based hydraulic diameter (13).

$$V_{\text{core}} = A_{\text{hot}} \left[t_{\text{hot}} + \frac{d_{v,\text{hot}}}{4} (1 + V_r) \right] \quad (16)$$

3.1.3. Quantities that scale with frontal area For a given heat transfer area and core volume, the choice of frontal area is a key driver of the velocity of the fluids. For an aligned flow heat exchanger (counterflow or coflow) of flow length $L = L_{\text{cold}} = L_{\text{hot}}$, the frontal area or the core A_{fr} can be broken down into the free-flow areas of the two fluids (assumed to occur at the same location) and metal area $A_{o,\text{metal}}$ at the free-flow cross section.

$$A_{\text{fr}} = A_{o,\text{cold}} + A_{o,\text{hot}} + A_{o,\text{metal}} \quad (17)$$

In heat exchangers with uniform cross section⁴, the metal area is related to the metal volume by multiplying by the flow length $A_{o,\text{metal}} \cdot L_{\text{hot}} = V_{\text{metal}}$. Equation (14) can then be used to obtain $A_{o,\text{metal}} = t_{\text{hot}} \cdot A_{\text{hot}}/L_{\text{hot}}$. Combining this in (17) with the heat transfer area from (14), the free-flow area based hydraulic diameter (13) as well as the free flow area ratio σ_r defined in (12):

$$A_{\text{fr}} = A_{o,\text{hot}} \left(1 + \sigma_r + \frac{4t_{\text{hot}}}{d_{h,\text{hot}}} \right) \quad (18)$$

In crossflow HEx there is a coupling between the frontal area of one side, and the flow length of the other side. This requires an additional parameter of how the two are coupled and results in there being two coupled equations for the hot and cold side frontal and free-flow areas.

3.2. Surface Performance

Heat transfer and pressure drop can be decomposed into first-order scaling and dimensionless coefficients that capture higher-order variations between surface types. The coefficient used for pressure drop is the friction factor f as defined by Kays and London [10], which reduces to the Fanning friction factor for cylindrical tubes but is more general in combining viscous shear and pressure forces.

Various analogies relate heat and momentum transfer, which rely on similar physical mechanisms. The Reynolds analogy states that for Prandtl number $\text{Pr} \approx 1$, the Stanton number $\text{St}/f \approx 0.5$. A generalisation of this over a larger range of Pr is the Chilton-Colburn analogy based on the Colburn factor $j \triangleq \text{StPr}^{2/3}$: $j/f \approx 0.5$. In practice, real heat exchanger surfaces do not achieve such high heat transfer coefficients and $j/f < 0.5$ is typical. Throughout this work, it will be assumed that:

$$\frac{j}{f} = 0.32 \quad \text{for } \text{Pr} = 0.7 \quad \Longleftrightarrow \quad \frac{\text{St}}{f} = 0.40 \quad (19)$$

This can be seen as the case for laminar flow in smooth circular tubes⁵ or surfaces with flow interruptions 30-42 times the hydraulic diameter using the general correlation derived by Miltén et al. using Kays and London experimental data [9, 10]. Furthermore, the Stanton number is assumed to have the following proportionality with the Reynolds number⁶, also from Miltén et al. [9]:

$$\text{St} \propto \text{Re}^{-0.413} \quad (20)$$

3.3. Governing Equations

The geometry and correlations are tied together to obtain the pressure drop for each side $i \in \{\text{hot, cold}\}$ as follows:

$$G_i = \frac{\dot{m}_i}{A_{o,i}} \quad (21)$$

⁴Assumed in this work. Not the case for flow over tube banks where the cross section varies.

⁵Constant heat flux boundary condition has $\text{St}/f = 4.36/16/0.7 = 0.39$.

⁶see below, (22) for definition

$$\text{Re}_i = \frac{G_i d_{h,i}}{\mu_i} \quad (22)$$

$$\Delta p_i \approx \frac{f_i A_i}{A_{o,i}} \cdot \frac{G_i^2}{2\rho_i} \quad (23)$$

$$(24)$$

where G_i is the mass velocity at the minimum free-flow area, Δp_i is the pressure drop (neglecting the entry, exit and expansion terms), and ρ_i the density for side i ⁷.

The mean thermal conductance $(hA)_i$ of side i can be determined from the definition of the Stanton number and the heat capacity flow rate $C_i = \dot{m}_i c_{p,i}$:

$$(hA)_i = \text{St}_i \cdot G_i \cdot c_{p,i} = \text{St}_i \cdot C_i \cdot \frac{A_i}{A_{o,i}} \quad (25)$$

The overall heat transfer performance or number of (heat) transfer units N_{tu} , assuming negligible wall thermal resistance and ideal fin efficiencies, is then given by ($C_{\min} \triangleq \min_i(C_i)$):

$$\frac{1}{N_{tu}} = \frac{A_{o,hot}}{A_{hot}} \cdot \frac{C_{\min}}{C_{hot}} \cdot \frac{1}{\text{St}_{hot}} \left(1 + \frac{C_{hot}}{C_{cold}} \cdot \frac{\text{St}_{hot}}{\text{St}_{cold}} \cdot \frac{\sigma_r}{A_r} \right) \quad (26)$$

3.4. Effectiveness–Number of Transfer Units Relations

For a counterflow heat exchanger, the thermal effectiveness ε is defined as the ratio of actual heat transfer to the maximum possible heat transfer. The traditional expression relating it to N_{tu} for constant heat capacity rates C_i only depends on their ratio [10] and can be rewritten as follows, explicitly referring to the hot and cold side heat capacity flow rates C_{hot} and C_{cold} ⁸:

$$\varepsilon = \frac{C_{hot}}{C_{\min}} \cdot \frac{e^{\Gamma} - 1}{e^{\Gamma} - C_{hot}/C_{cold}} \quad (27)$$

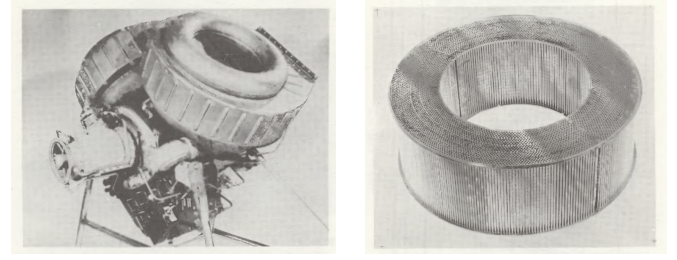
where

$$\Gamma \triangleq N_{tu} \cdot \frac{C_{\min}}{C_{hot}} \left(1 - \frac{C_{hot}}{C_{cold}} \right) \quad (28)$$

4. APPLICATION

In this section, we apply the frameworks of this paper to inform a design of a real aviation performance enhancing heat exchanger, namely the recuperator of a helicopter turboshaft engine as shown in Fig. 4 and detailed in Table 2. McDonald retrofitted a 313 kW engine with a bolt-on tubular crossflow recuperator that conducted several test flights [8]. They also designed a second heat exchanger for a redesigned engine.

The recorded performance of these heat exchangers has been used to infer a reference heat exchanger design that will be used for the remainder of this paper to demonstrate the utility of the practical availability framework to a designer in a real application. This reference point assumes a heat exchanger effectiveness of 60%, a hot side pressure drop of 10.8%, a cold side pressure



(a) Recuperated Engine

(b) Heat Exchanger

FIGURE 4: Bolt-on recuperator for helicopter engine [8]

TABLE 2: Helicopter engine design parameters

Engine mass ⁹	72 kg
Shaft power	313 kW
Mass flow rate ($\dot{m}_i$)	1.9 kg/s
Shaft efficiency ¹⁰ (η_{th})	25%
Heat Exchanger	
Tube mass (m_{core})	13.3 kg
Hot inlet:	980 K, ~1 bar
Cold inlet:	576 K, 7.2 bar
Diameter (outer/inner)	0.25 m, 0.15 m
Tube length	6.3 cm

drop of 1.2%, and a hot side dimensionless mass velocity of 2×10^{-2} (obtained by using the geometry and mass flow shown in Table 2). The ratio of hydraulic diameters $d_{h,cold}/d_{h,hot} = 0.44$ was chosen as it minimised the sum of both relative pressure drops. Note that these pressure drops are slightly higher than the combined 10% achieved by the real bolt-on heat exchanger and the dimensionless mass velocity is four times lower. This was done to achieve a reference general counterflow heat exchanger using the methodology of section 3 of similar performance to the real crossflow heat exchanger as presented by McDonald [8].

Three design choices will be considered in the following three subsections, using the frameworks presented previously. These design choices will involve different trade-offs between heat transfer and pressure drop to improve upon the helicopter recuperator's reference performance. Firstly, improvements to the design will be considered at fixed heat exchanger core mass and volume. As the volume being fixed, the product of frontal area and length is also fixed, but the frontal area (and thus the mass velocity) can be changed if the length is appropriately adjusted to keep the volume constant. Then, the variation of the volume and mass velocity will be considered together for a fixed core mass. This assumes that the hydraulic diameter or compactness can be varied independently of the wall thickness (hence the constant mass assumption is equivalent to constant heat transfer area). Lastly, for a given hydraulic diameter, the variation of mass velocity and heat exchanger core mass (now again assumed proportional to core volume) will be studied in combination with a basic engine model.

⁷The density can change substantially due to heat transfer. This is neglected in this work for simplicity.

⁸The expressions is still symmetric with respect to two heat capacity flow rates, which can be seen by multiplying numerator and denominator by $-e^{-\Gamma}$.

⁹unrecuperated

¹⁰recuperated

4.1. Varying mass velocity (fixed mass and volume)

The reference design has a certain heat exchanger core mass and volume which for this section will not be changed. The dimensionless mass velocity can be changed by varying the frontal area and adjusting the length such that the volume is kept constant. At lower mass velocities the Reynolds number decreases, resulting in an increase in friction coefficient and Stanton number. For the fixed heat transfer area therefore, lower mass velocities mean higher NTU and higher pressure drop.

As can be seen in Fig. 5a the pressure drop now increases non-linearly with NTU.

$$\Delta p/p_{in} \propto N_{tu}^{4.41} \quad (29)$$

Where the numerical value of the exponent depends on the adequate correlations for Stanton number and friction coefficient.

Neither the conventional method nor the classical availability method shown in Fig. 5b are helpful in deciding which mass velocity is best for this fixed heat exchanger core mass and volume.

Figure 5c on the other hand, where the thermal creation and viscous dissipation terms of equation (7) have been separated out, clearly shows that a slightly lower mass velocity: from $g_{hot}^2 = 2 \times 10^{-3}$ at the reference design to $g_{hot}^2 = 9.4 \times 10^{-3}$ at the optimum for the degrees of freedom considered in this section.

Coming back to Fig. 5a, it is not necessarily obvious that for this design choice, going from an effectiveness of 60% and a combined pressure drop of 12.0% to an effectiveness of 54% and a combined pressure drop of 4.5% is better. But if these values are entered into equation (9) and (7), the latter choice is better for all fluid pressure ratios and any practical temperature ratios where recuperation would be considered in practice.

For this first design choice, the conventional and classical availability methods have been shown to be of little use to the designer, while the practical availability method guides the designer towards the right balance of frontal area and length for a given volume. Since the subsequent two design choices will also require finding the optimal mass velocity for fixed mass and volume, only the practical availability method will be used.

4.2. Varying mass velocity and volume (fixed mass)

In another scenario, a heat exchanger designer might be allocated a fixed volume and mass within which to fit a best heat exchanger. But in practice there may be ways to find more volume if the associated benefit were worth it. Therefore, we now consider the potential benefit of changing the volume while keeping the mass fixed. Additionally, the wall thickness will be assumed constant (which can be thought of as some minimum manufacturable thickness). The assumptions of fixed mass and thickness combine to also fixing the heat transfer area. Varying the volume for a fixed heat transfer area is tantamount to varying the hydraulic diameter as can be seen in equation (16), which is equivalent to changing the compactness of the heat exchanger.

The assumption of fixed wall thickness decouples the two typical benefits of high compactness and low hydraulic diameters in heat exchangers:

1. Compact heat exchangers (i.e. with small hydraulic diameters) have on the one hand the benefit of structural strength,

meaning that for a given fluid pressure difference, smaller wall thicknesses are required. This results in having more heat transfer area for a fixed core mass. This benefit is not considered in this section as the wall thickness is assumed constant and independent of hydraulic diameter.

2. Compact heat exchangers also take up less volume for a given heat transfer area. This is beneficial if volume is considered fixed or constrained. In this study however we are neglecting the penalties of volume (drag and potentially more structural mass needed to enclose the additional volume), and instead are just interested in the impact on pressure drop and heat transfer.

First of all, assume we are at a fixed mass velocity (and hence frontal area). Increasing volume by increasing the hydraulic diameter is therefore independent of the heat transfer and free-flow areas. It does however impact the Reynolds number and therefore the friction coefficient and Stanton number. Larger volumes will therefore have less pressure drop and less heat transfer, as the area is fixed. The practical availability framework would allow us to determine the best volume at a given mass velocity – in Fig. 6 this will be equivalent to finding the optimum along a horizontal line.

As has been shown in the previous section, for each given volume, there also is an optimal choice of mass velocity. Combining the two results in the contour plot shown in Fig. 6. The line of optimum mass velocity for a given volume can be seen to increase with increasing volume: at larger volumes (from larger hydraulic diameters), it is worthwhile to elongate the flow volumes more to increase the mass velocity and hence maintain an optimal pressure drop and heat transfer.

In fact, for the specific case of this heat exchanger, it was found that along the line of optimum mass velocity and core volume, the pressure drop remains roughly constant while the heat transfer increases with volume.

In conclusion, if it were required to have a fixed wall thickness independent of hydraulic diameter, and there were no penalty to additional volume (slow flying aircraft, or aircraft with spare volume) but a fixed mass nonetheless, it would be beneficial to go to higher hydraulic diameters and adjust the length such that the flow is at higher mass velocities to achieve the best increase in practical availability.

4.3. Optimising mass velocity, volume, and mass (unconstrained)

Now it is assumed again that the reference diameter is as small as can be durably integrated into an aviation heat exchanger. The previous two sections have considered fixed mass of heat exchanger. For a given mass and hydraulic diameter, the optimal mass velocity (i.e. free flow area) can be chosen such that the benefits of heat transfer and penalty of pressure drop are balanced in terms of the practical availability (see section 4.1).

This will now be extended to different heat exchanger core masses. The key trade-off with heat exchanger mass (assuming negligible drag penalty), is that it saves fuel, but adds a fixed mass that must be carried throughout the mission. The benefit of the heat exchanger will therefore be heavily dependent on the typical

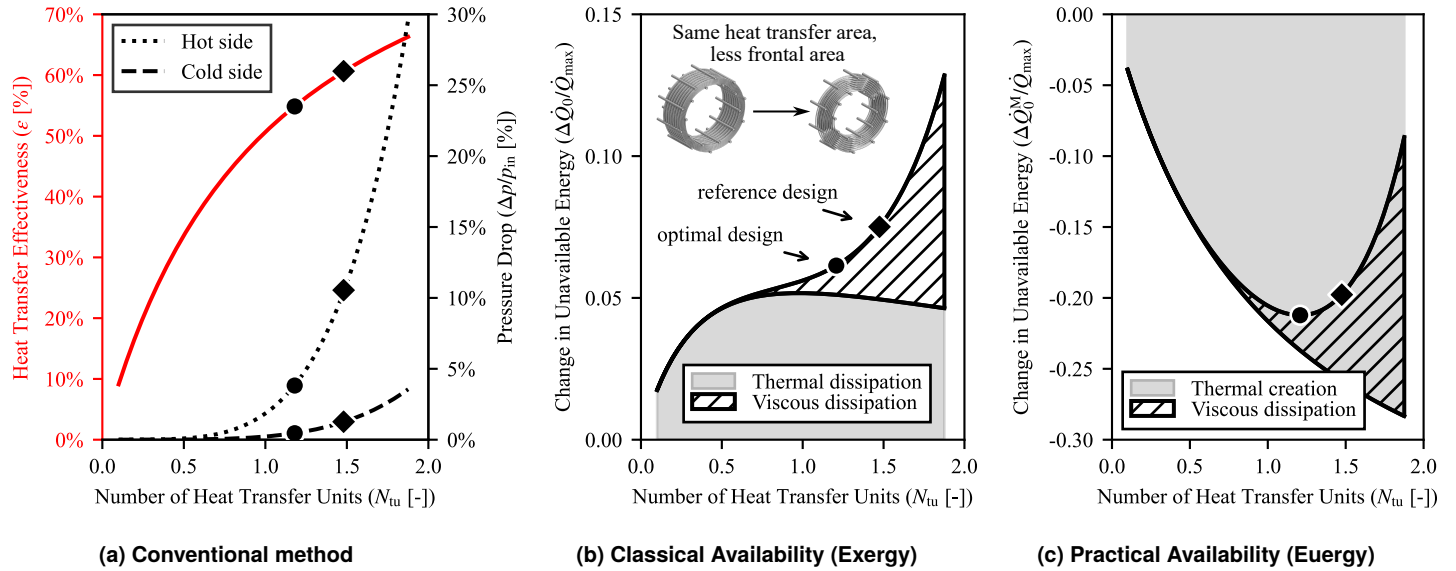


FIGURE 5: A comparison of three frameworks for a heat exchanger as the mass velocity is varied. The mass and the volume of the heat exchanger is fixed. The mass velocity is increased by increasing the frontal area while reducing the heat exchanger length to maintain volume.¹¹

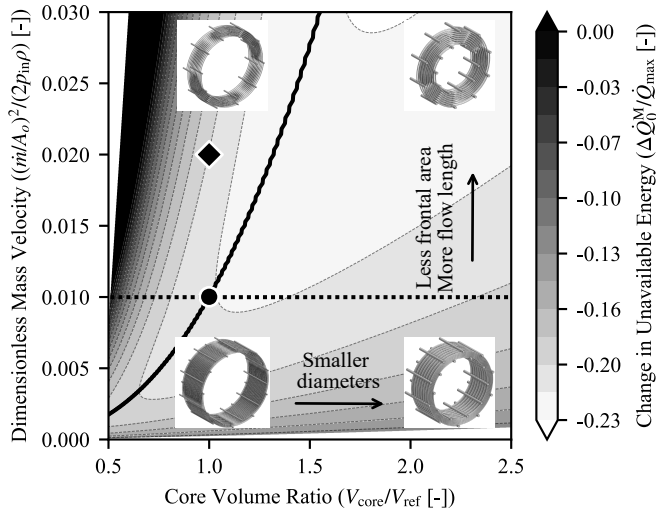


FIGURE 6: Practical availability analysis for heat exchangers of fixed mass but varying volume and mass velocity.

duration τ_{mission} (taken to be 2h), as well as the thermal efficiency of the engine (which will change with different heat exchanger sizes, but for simplicity an intermediate constant value of 25% has been chosen) and the specific heat content of the fuel (12 kWh/kg for kerosene). The simple cycle model to demonstrate the trade-off in this section assumes that for every unit of practical unavailable energy made available again by the heat exchanger, a real turbine converts this into actual work (at an efficiency assumed at 80%) and avoids an equivalent mass of fuel as per the above mentioned efficiency.

$$\Delta m_{\text{fuel}} = \frac{\Delta Q_0^M}{\dot{Q}_{\text{max}}} \cdot \dot{m} c_p (T_{\text{hot,in}} - T_{\text{cold,in}}) \cdot \frac{\eta_{\text{turb}}}{\eta_{\text{th}}} \cdot \frac{\tau_{\text{mission}}}{\text{LHV}_{\text{fuel}}} \quad (30)$$

Therefore Fig. 7 shows that as more heat exchanger mass is added, if the optimum mass velocity is chosen by adjusting the mass velocity as in section 4.1, the fuel saving continually increases. The combined impact on take-off mass of fuel mass saving and heat exchanger fixed mass leads to an optimum.

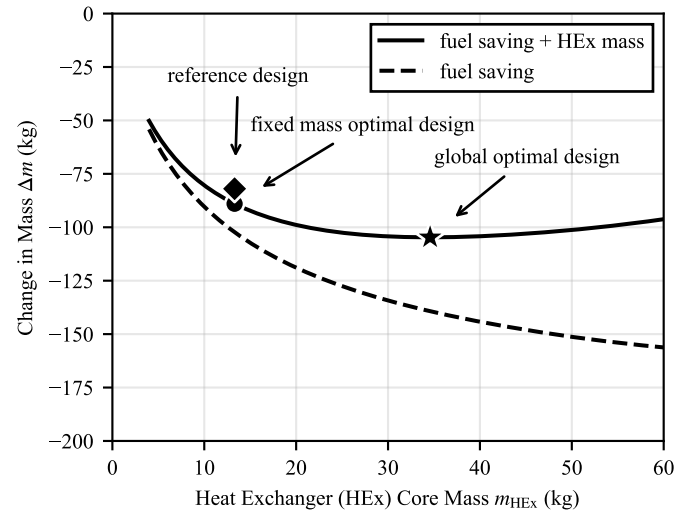


FIGURE 7: Mass saving for different helicopter recuperator sizes (mass and volume). The mass velocity is optimized for each size using the practical framework.

¹¹At the reference point $g_{\text{hot}}^2 = 20 \times 10^{-3}$. The plots assume for all other points $C_{\text{cold}} = 0.95 C_{\text{hot}}$, $\text{St}/f = 0.4$ for both fluids, equal friction factors $f_{\text{hot}} = f_{\text{cold}}$, hydraulic diameters $d_{\text{h,cold}} = 0.44 d_{\text{h,hot}}$, temperature ratio $T_{\text{hot,in}} / T_{\text{cold,in}} = 1.7$, pressure ratio $p_{\text{cold,in}} / p_{\text{hot,in}} = 7.2$, $T_{\text{cold,in}} / T_{\text{dead}} = 1.9$, $p_{\text{cold,in}} / p_{\text{dead}} = 1.03$, both fluids are ideal gases with constant specific heats and similar molar mass, and heat transfer areas are equal $A_{\text{hot}} = A_{\text{cold}}$

A higher fidelity model would account for the volume penalty, the eventual resizing of cycle and the increased thermal efficiency due to recuperation, all three of which would probably reduce the fuel mass savings at higher heat exchanger masses, pushing the optimum to lower masses of heat exchanger. However, this simple model is still useful to demonstrate that mass must be the primary focus of an aviation heat exchanger design. The utility of practical availability in this context was to capture the true value of heat and hence inform design choices involving trade-offs between heat transfer and pressure drop. Although no local metric can capture all the impacts of a real cycle, using the correct ideal machine for applications where bottoming cycles downstream are not an option captures the correct physical intuition.

5. CONCLUSION

Three frameworks for comparing heat transfer and pressure drop in heat exchanger design have been considered. The classical availability method (exergy), albeit well studied in literature, was shown to mislead design, especially in high mass velocity cases. Therefore, the conventional method considers heat transfer and pressure drop as two metrics that cannot be traded in general. The practical availability method (euergy) was applied in detail to cases where the other methods could not inform design. The design of fuel burn reducing aviation heat exchangers was of particular interest as the mass and volume constraints involved require multiple design choices that trade heat transfer and pressure drop. The usefulness of the practical availability method was demonstrated by considering three such choices to a generic helicopter recuperator based on a high-fidelity design from literature.

- For a fixed mass and volume, an optimum mass velocity exists achievable by changing the frontal area and adjusting the length to conserve volume.
- This optimum mass velocity can also be found at different volumes but fixed masses by varying the hydraulic diameter and assuming a constant wall thickness. Larger volumes are always worthwhile in terms of practical availability, even if the heat transfer area and mass cannot be increased. Which is a helpful insight to be traded with the penalties of additional volume that are outside of the scope of this work.
- The optimum mass velocity can also be found for different heat transfer areas and hence mass and volume, by assuming a fixed hydraulic diameter. A rudimentary first order model of the penalty of added mass to the design mission was applied to find the global optimum improved recuperator design, assuming negligible drag penalty (reasonable for a helicopter).

The authors hope that through this demonstration that the practical availability framework was shown to be an informative tool for heat exchanger design where trade-offs between heat transfer and pressure drop are necessary.

ACKNOWLEDGMENTS

The authors acknowledge the financial support of the Engineering and Physical Sciences Research Council (EPSRC) and the Center for Doctoral Training in Future Propulsion and Power.

NOMENCLATURE

Roman letters

A	Heat transfer surface area [m ²]
A_{fr}	Frontal area [m ²]
A_o	Minimum free-flow area [m ²]
A_r	Heat transfer area ratio [-]
C	Heat capacity rate, $\dot{m}c_p$ [W K ⁻¹]
C_r	Heat capacity ratio, C_{min}/C_{max} [-]
c_p	Specific heat capacity [J kg ⁻¹ K ⁻¹]
d_h	Hydraulic diameter [m]
f	Equivalent friction factor [-]
G	Mass velocity, $\dot{m}/A_o$ [kg m ⁻² s ⁻¹]
g^2	Mass velocity parameter [-]
h	Specific enthalpy [J kg ⁻¹]
j	Colburn j -factor, $StPr^{2/3}$ [-]
k	Isentropic exponent, $(\gamma - 1)/\gamma$ [-]
L	Heat exchanger flow length [m]
l_s	Flow interruption spacing [m]
m	Mass [kg]
$\dot{m}$	Mass flow rate [kg s ⁻¹]
N_{tu}	Number of transfer units [-]
p	Pressure [Pa]
$\dot{P}$	Pumping power [W]
$\dot{Q}$	Heat transfer rate [W]
s	Specific entropy [J kg ⁻¹ K ⁻¹]
$\dot{S}$	Entropy rate [W K ⁻¹]
T	Temperature [K]
t	Wall thickness [m]
V	Volume [m ³]
$\dot{W}_0$	Available work potential (exergy) [W]
$\dot{W}_0^M$	Mechanical work potential (euergy) [W]

Greek letters

β_i	Surface area density A_i/V_i [m ² /m ³]
γ	Ratio of specific heats [-]
ε	Thermal effectiveness [-]
ε_W	Work potential effectiveness [-]
η	Efficiency [-]
ρ	Density [kg m ⁻³]
σ_r	Free-flow area ratio [-]
τ	Mission duration [s]

Dimensionless groups

Pr	Prandtl number
Re	Reynolds number, Gd_h/μ
St	Stanton number, $h_{conv}/(Gc_p)$

Subscripts

bulk	Bulk fluid
c / cold	Cold stream
d	Dead state
e	Isentropic expansion exit
h / hot	Hot stream
i	Stream index ($i \in \{h, c\}$)

in	Inlet
out	Outlet
ov	Overall
t	Stagnation / total
wall	Wall surface

REFERENCES

- [1] Keenan, Joseph H. “Availability and irreversibility in thermodynamics.” *British Journal of Applied Physics* Vol. 2 No. 7 (1951): p. 183. DOI [10.1088/0508-3443/2/7/302](https://doi.org/10.1088/0508-3443/2/7/302). URL <https://iopscience.iop.org/article/10.1088/0508-3443/2/7/302>.
- [2] Miller, R. J. “Mechanical Work Potential.” *Proceedings of the ASME Turbo Expo 2013: Turbine Technical Conference and Exposition. Volume 6A: Turbomachinery*: p. V06AT36A032. 2013. ASME, San Antonio, Texas, USA. DOI [10.1115/GT2013-95488](https://doi.org/10.1115/GT2013-95488).
- [3] Jardine, L. and Miller, R. “The Effect of Heat Transfer on Turbine Performance.” *ASME Journal of Turbomachinery* (2021) DOI [10.1115/1.4051109](https://doi.org/10.1115/1.4051109).
- [4] Bejan, Adrian. *Advanced Engineering Thermodynamics*, 4th ed. John Wiley & Sons Inc., Hoboken, New Jersey (2016).
- [5] Navaie, Ali Rezaie. “Thermal Design and Optimization of Heat Recovery Steam Generators and Waste Heat Boilers.” Ph.d. thesis, Technische Universität Berlin, Berlin. 2017. URL <https://d-nb.info/1156187052/34>.
- [6] Sampath, Suresh. “Fault diagnostics for advanced cycle marine gas turbine using genetic algorithm.” Phd thesis, Cranfield University. 2003. URL <http://dspace.lib.cranfield.ac.uk/handle/1826/10204>.
- [7] Brewer, G. Daniel. *Gas Turbine Technology*. CRC Press, Boca Raton, FL (1991).
- [8] McDonald, C. F. “Gas Turbine Recuperator Technology Advancements.” *Proceedings of the ASME 1972 International Gas Turbine and Fluids Engineering Conference and Products Show*: p. V001T01A031. 1972. ASME, San Francisco, California, USA. DOI [10.1115/72-GT-32](https://doi.org/10.1115/72-GT-32).
- [9] Miltén, P., Johnsson, I., Lundblad, A. and Xisto, C. “Generalized Method for the Conceptual Design of Compact Heat Exchangers.” *ASME Journal of Engineering for Gas Turbines and Power* Vol. 146 No. 11 (2024): p. 111018. DOI [10.1115/1.4065922](https://doi.org/10.1115/1.4065922).
- [10] Kays, W. M. and London, A. L. *Compact Heat Exchangers*, 3rd ed. Scientific International (2018). Book originally published in 1984.
- [11] Zhang, C. and Gümmer, V. “The potential of helicopter turboshaft engines incorporating highly effective recuperators under various flight conditions.” *Aerospace Science and Technology* Vol. 88 (2019): pp. 84–94. DOI [10.1016/j.ast.2019.03.008](https://doi.org/10.1016/j.ast.2019.03.008).
- [12] LaHaye, P. G., Neugebauer, F. J. and Sakhuja, R. K. “A Generalized Prediction of Heat Transfer Surfaces.” *ASME Journal of Heat Transfer* Vol. 96 No. 4 (1974): pp. 511–517. DOI [10.1115/1.3450237](https://doi.org/10.1115/1.3450237).
- [13] Roth, Bryce. “A work potential perspective of engine component performance.” *37th Joint Propulsion Conference and Exhibit*. 2001. DOI [10.2514/6.2001-3300](https://doi.org/10.2514/6.2001-3300).
- [14] Nichols, J. B. “An Energy Basis for Comparison of Performance of Combustion Chambers.” *Trans. ASME* Vol. 75 No. 1 (1953): pp. 29–33. DOI [10.1115/1.4015200](https://doi.org/10.1115/1.4015200). URL <https://doi.org/10.1115/1.4015200>. Originally published January 1953; online as of July 29, 2022.
- [15] Pratt & Whitney. “Alternative Fuel.” (2024). Accessed 2025-12-08, URL <https://www.prattwhitney.com/en/future-of-flight/alternative-fuel>.
- [16] Ostrower, Jon. “CFM RISE Program Concept.” (2021). Accessed 2025-12-08, URL <https://x.com/jonostrower/status/1404430361908502532/photo/4>.
- [17] FlightGlobal. “Water-Enhanced Turbofan: Revolutionary Propulsion Concept Based on a Gas Turbine Engine.” (2022). Accessed 2025-12-08, URL <https://tinyurl.com/WETmtu>.
- [18] Encyclopedia Britannica. “Entropy.” (2024). Accessed 2025-12-08, URL <https://www.britannica.com/science/entropy-physics>. “[Entropy is] a measure of the energy that is unavailable for useful work”.

APPENDIX A. APPENDIX: THRESHOLD G2 FOR MINIMUM ENTROPY GENERATION

For the specific problem considered in this section, namely determining the best N_{tu} by lengthening a balanced counterflow heat exchanger, in the classical availability framework

Assume hot pressure drop is equal in magnitude or greater than cold pressure drop, then let us focus on the condition for the hot side mass velocity. Also assumes balanced counterflow heat exchanger so no point in writing for C_h and C_c ! Especially since for H2 it is known that it doesn't work.

The existence of a local minimum in entropy generation depends on the magnitude of g^2 . It can be shown that for a variation in heat transfer area A , no minimum in entropy generation occurs in a balanced heat exchanger if:

$$g_{hot}^2 \cdot \frac{St_{hot}}{f_{hot}} \left[1 + \frac{St_{hot}}{St_{cold}} \frac{\sigma_r}{A_r} \right] > K \left(\frac{T_{hot,in}}{T_{cold,in}} - 1 \right) \quad (31)$$

where St_{ov} is the overall Stanton number (see Eq. XX) and f_i is the friction factor on side i . The constant $K \approx 0.05$ for heat exchangers with $g_{hot}^2 \approx g_{cold}^2$, and $K \approx 0.03$ when $g_{hot}^2 \gg g_{cold}^2$ (within 10%). The derivation assumes a heat exchanger with identical fluids and mass flow rates.

The critical condition is determined by the maximum value of g_i^2 between the two sides. For all recuperators in Table 1, this is the hot-side value, where the lower density results in higher velocities for a given mass flux.